

SHRUTI AKKALA

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SUMMARY

Full-Stack Software Engineer with 6+ years of experience building scalable customer-facing platforms and distributed systems using React, TypeScript, Java, Spring Boot, and AWS. Experienced in architecting end-to-end products across frontend UI, backend microservices, cloud infrastructure, and large-scale Spark data pipelines serving millions of users.

WORK EXPERIENCE

Capital One (Formerly Discover Financial Services) - Chicago, IL

Jan 2023 – Present

Senior Software Engineer - Authentication, Biometrics & Identity Platforms

- Developed reusable **React/TypeScript** UI components with asynchronous state management for biometric authentication and push notification workflows, integrating with Java Spring Boot APIs to deliver a seamless passwordless experience for millions of users.
- Designed and built a new Multi-Factor Authentication (MFA) capability, integrating FIDO protocols into distributed **Java backend microservices** to enable high-volume, biometric authentication for Discover's mobile experience.
- Migrated biometric auth services from on-prem to AWS, refactoring legacy components into cloud-native **Spring Boot microservices** — improving deployment scalability and reducing infrastructure overhead.
- Reduced **p95 latency by 78%** (1.13s to 252ms) and backend production errors by 91% through performance tuning, Kubernetes deployments, and observability via Datadog.
- Built and owned a Spark + Java distributed pipeline processing 45M debit card transaction records daily (~150 GB), meeting a hard 4:30 AM financial SLA — replacing a legacy IBM mainframe and reducing **infrastructure costs by \$2.5M**.

Verifone - Chicago, IL

Sep 2019 – Jan 2023

Software Engineer - Devices and Fleet Management

- Developed **Python and Bash** automation for build, packaging, and deployment workflows eliminating manual operational processes and reducing backend operational costs by **23%**.
- Implemented package validation, versioning, and device compatibility rules to ensure software distribution by model and release requirements.
- Integrated cloud orchestration with Linux-based terminal update workflows to **support real-time rollouts** and **automated rollbacks**.
- Owned production issue resolution via ServiceNow by reproducing customer terminal environments and performing **RCA** across device software, network conditions, and backend regressions; delivered fixes or escalation-ready RCA to partner teams.

Amazon – Seattle, WA

Sep 2018 – Sep 2019

Software Engineer - Amazon Business Personalization

- Designed and implemented high-scale Java-based microservices, supporting **10,000 TPS with p99 latency of 50ms** through performance tuning and horizontal scalability.
- Built the “Customers also bought” carousel for the Amazon Business homepage, building backend services and APIs to power dynamic product recommendations.
- Delivered production-grade code with 95%-unit test coverage and 92% branch coverage, ensuring high reliability for a high-throughput microservice handling **100K+ daily transactions**.
- Maintained operational excellence by leading root cause analysis for on-call incidents and improving system reliability and availability for business-critical services.

SKILLS

- **Languages & Frameworks:** Java, Spring Boot, Python, JavaScript, TypeScript, React, Node.js
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda), Kubernetes, Docker, Git, CI/CD (Jenkins)
- **Databases:** SQL, PostgreSQL, Oracle, MySQL, MongoDB
- **Testing & Observability:** JUnit, Mockito, Datadog
- **AI-Assisted Development:** GitHub Copilot, Claude Code, RAG workflows, AI-assisted debugging and code review

EDUCATION

Illinois Institute of Technology, Chicago, IL, Master of Science in Computer Science

May 2018